

B3: Atomic & Laser Physics — Model Solutions, Trinity 2024

Question 1 — Spin–orbit interaction and fine structure

Problem. One-electron atom: derive H_{SO} (Thomas factor via $g_s \rightarrow g_s - 1$), show the $l \neq 0$ splitting $\Delta E = Z^4 \alpha^2 \hbar c R_\infty / [n^3 l(l+1)]$, then apply to H ($2 \rightarrow 3$) vs. He^+ ($4 \rightarrow 6$): fine structure, mass shift, E1 selection rules, Doppler width, and the Z giving fine structure equal to the $2 \rightarrow 3$ energy of H.

(a) Origin and derivation of H_{SO}

In the electron rest frame the orbiting nucleus produces a magnetic field that couples to the electron spin moment $\boldsymbol{\mu}_s = -g_s \mu_B \mathbf{S} / \hbar$.

Lorentz boost of the nuclear $\mathbf{E} = -\nabla\phi$ to the electron frame, $\mathcal{O}(v^2/c^2)$:

$$\mathbf{B}' = -\frac{1}{c^2} \mathbf{v} \times \mathbf{E}.$$

Central potential, $\mathbf{E} = -\hat{\mathbf{r}} \partial_r \phi$:

$$\mathbf{v} \times \mathbf{E} = \frac{1}{r} \partial_r \phi (\mathbf{r} \times \mathbf{v}).$$

Identify $\mathbf{L} = m_e \mathbf{r} \times \mathbf{v}$:

$$\mathbf{B}' = \frac{1}{m_e c^2} \frac{1}{r} \partial_r \phi \mathbf{L}.$$

Zeeman energy $-\boldsymbol{\mu}_s \cdot \mathbf{B}'$; substitute Thomas-corrected $g_s - 1 \approx 1$ and $U = -e\phi$:

$$\boxed{H_{\text{SO}} = \frac{1}{m_e^2 c^2} \frac{1}{r} \frac{\partial U}{\partial r} \mathbf{L} \cdot \mathbf{S}.} \quad (1)$$

Hydrogenic $U = -Ze^2/(4\pi\epsilon_0 r)$:

$$H_{\text{SO}} = \frac{Ze^2}{4\pi\epsilon_0 m_e^2 c^2 r^3} \mathbf{L} \cdot \mathbf{S} \equiv \xi(r) \mathbf{L} \cdot \mathbf{S}. \quad (2)$$

(b) Splitting of a configuration into two levels

Coupled basis $\mathbf{J} = \mathbf{L} + \mathbf{S}$:

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2} (\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2).$$

For $s = 1/2$, $l \neq 0$, $j = l \pm 1/2$:

$$\langle \mathbf{L} \cdot \mathbf{S} \rangle_{j=l+1/2} = \frac{\hbar^2}{2} l, \quad \langle \mathbf{L} \cdot \mathbf{S} \rangle_{j=l-1/2} = -\frac{\hbar^2}{2} (l+1).$$

Splitting $\Delta E_j = \langle \xi(r) \rangle \langle \mathbf{L} \cdot \mathbf{S} \rangle_j$:

$$\Delta E = \langle \xi(r) \rangle \frac{\hbar^2}{2} (2l+1).$$

Insert hydrogenic $\langle 1/r^3 \rangle = 2(Z/na_0)^3/[l(l+1)(2l+1)]$:

$$\Delta E = \frac{Ze^2\hbar^2}{4\pi\epsilon_0 m_e^2 c^2} \frac{Z^3}{n^3 a_0^3} \frac{1}{l(l+1)}.$$

Apply $\alpha = e^2/(4\pi\epsilon_0\hbar c)$, $a_0 = \hbar/(\alpha m_e c)$, $hcR_\infty = \frac{1}{2}m_e c^2 \alpha^2$:

$$\boxed{\Delta E = \frac{Z^4 \alpha^2 hcR_\infty}{n^3 l(l+1)} \quad (l \neq 0).} \quad (3)$$

(c) H ($2 \rightarrow 3$) vs. He⁺ ($4 \rightarrow 6$)

Bohr energy $E_{n \rightarrow n'} = hcR_\infty Z^2(1/n^2 - 1/n'^2)$.

H ($Z = 1$, $2 \rightarrow 3$): $hcR_\infty(1/4 - 1/9) = \frac{5}{36}hcR_\infty$.

He⁺ ($Z = 2$, $4 \rightarrow 6$): $4hcR_\infty(1/16 - 1/36) = \frac{5}{36}hcR_\infty$.

Same Bohr wavelength $\lambda \simeq 656$ nm (H $_{\alpha}$), $\nu_0 \simeq 4.57 \times 10^{14}$ Hz.

Largest fine-structure splitting is at smallest $l \neq 0$, i.e. $l = 1$. Apply (3):

$$\Delta E_{n=2,l=1}^{\text{H}} = \frac{1 \cdot \alpha^2}{8 \cdot 2} hcR_\infty = \frac{\alpha^2}{16} hcR_\infty.$$

$$\Delta E_{n=4,l=1}^{\text{He}^+} = \frac{16 \alpha^2}{64 \cdot 2} hcR_\infty = \frac{\alpha^2}{8} hcR_\infty.$$

Express as fraction of $h\nu_0 = \frac{5}{36}hcR_\infty$:

$$\boxed{\frac{\Delta\nu^{\text{H}}}{\nu_0} = \frac{9\alpha^2}{20} \simeq 2.4 \times 10^{-5}, \quad \frac{\Delta\nu^{\text{He}^+}}{\nu_0} = \frac{9\alpha^2}{10} \simeq 4.8 \times 10^{-5}.} \quad (4)$$

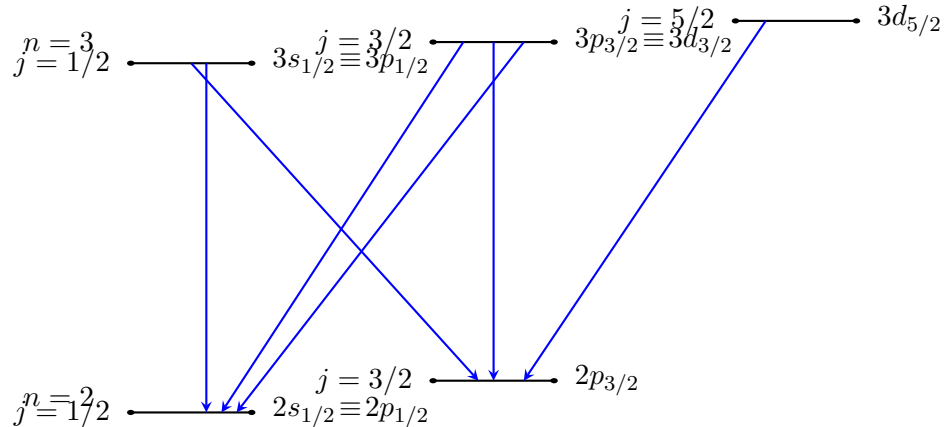
Mass shift. Reduced-mass Rydberg $R_M = R_\infty/(1 + m_e/M)$ gives

$$\frac{\Delta\nu_{\text{mass}}}{\nu_0} \simeq \frac{m_e}{M_p} - \frac{m_e}{M_\alpha} \simeq \frac{3}{4} \cdot \frac{1}{1836} \simeq 4 \times 10^{-4},$$

an order of magnitude larger than the fine-structure splittings; mass shift dominates the wavelength difference.

(d) Energy-level diagram and E1 selection rules, H $n = 2 \rightarrow 3$

Levels $n^{2S+1}L_J$, $S = 1/2$. Dirac-degeneracy gives $2s_{1/2} \equiv 2p_{1/2}$, $3s_{1/2} \equiv 3p_{1/2}$, $3p_{3/2} \equiv 3d_{3/2}$.



E1 selection rules:

$$\Delta l = \pm 1, \quad \Delta s = 0, \quad \Delta j = 0, \pm 1 \ (0 \not\rightarrow 0), \quad \Delta m_j = 0, \pm 1. \quad (5)$$

Justification of $\Delta l = \pm 1$. $\hat{\mathbf{d}} = -e\hat{\mathbf{r}}$ is a rank-1 odd-parity tensor. Parity selection forces $l - l'$ odd; Wigner–Eckart forces $|l - l'| \leq 1$. Together: $\Delta l = \pm 1$.

(e) Doppler broadening at $T = 300$ K

Thermal Doppler FWHM:

$$\frac{\Delta\nu_D}{\nu_0} = \sqrt{\frac{8k_B T \ln 2}{Mc^2}}.$$

Insert $k_B T \simeq 4.14 \times 10^{-21}$ J, $M = m_p$, $c^2 \simeq 9 \times 10^{16} \text{ m}^2\text{s}^{-2}$:

$$\frac{\Delta\nu_D^H}{\nu_0} = \sqrt{\frac{8 \cdot 4.14 \times 10^{-21} \cdot 0.693}{1.67 \times 10^{-27} \cdot 9 \times 10^{16}}}.$$

Evaluate:

$$\frac{\Delta\nu_D^H}{\nu_0} \simeq 1.2 \times 10^{-5}.$$

He^+ has $M = 4m_p$, so width scales by $1/\sqrt{4} = 1/2$:

$$\boxed{\frac{\Delta\nu_D^H}{\nu_0} \simeq 1.2 \times 10^{-5}, \quad \frac{\Delta\nu_D^{\text{He}^+}}{\nu_0} \simeq 6 \times 10^{-6}.} \quad (6)$$

Further mechanism in a discharge lamp. Pressure (collisional) broadening: random collisional dephasing of the radiating dipole, Lorentzian width $\sim 1/\tau_{\text{coll}}$. (Stark broadening from plasma microfields is also strong for H because of its linear Stark effect.)

(f) Fine structure equal to the H $2 \rightarrow 3$ energy

Equate $n = 2, l = 1$ splitting to $\frac{5}{36}hcR_\infty$:

$$\frac{Z^4 \alpha^2}{16} hcR_\infty = \frac{5}{36} hcR_\infty.$$

Solve for Z^4 :

$$Z^4 = \frac{80}{36 \alpha^2} \simeq 4.2 \times 10^4.$$

$$\boxed{Z \simeq 14 \text{ (e.g. Si}^{13+}\text{)}}. \quad (7)$$

Alkali Z-scaling. Valence np electron penetrates the unscreened core; SO coupling samples the bare nuclear field. Empirical ΔE_{FS} : Na 0.42 meV, K 2.0 meV, Rb 6.96 meV, Cs 28 meV — super-quartic in Z because both the inner field and the penetrating-wavefunction amplitude grow with Z .

Question 2 — Hyperfine structure and the hydrogen MASER

Problem. Hamiltonian $\mathcal{H} = A \mathbf{I} \cdot \mathbf{J} + g_J \mu_B \mathbf{J} \cdot \mathbf{B} - g_I \mu_N \mathbf{I} \cdot \mathbf{B}$. Identify each term, find the weak/strong-field crossover for ground-state H ($A/h = 1420$ MHz, $I = 1/2$), derive the eigenenergies in both regimes, and analyse a Stern–Gerlach beam with $B_z = 1$ T, $\partial_z B_z = 20$ T/m as the basis of an H maser.

(a) Origin of each term

$A\mathbf{I}\cdot\mathbf{J}$: hyperfine coupling of nuclear moment $\boldsymbol{\mu}_I = g_I\mu_N\mathbf{I}/\hbar$ to the electron-generated magnetic field at the nucleus (dominated by the Fermi-contact term for s states).

$g_J\mu_B\mathbf{J}\cdot\mathbf{B}$: electronic Zeeman, energy of $\boldsymbol{\mu}_J = -g_J\mu_B\mathbf{J}/\hbar$ in $\mathbf{B}$.

$-g_I\mu_N\mathbf{I}\cdot\mathbf{B}$: nuclear Zeeman, smaller by $m_e/m_p \sim 1/1836$.

Regimes. Weak field $g_J\mu_B B \ll A$: $\mathbf{I}, \mathbf{J}$ couple to $\mathbf{F} = \mathbf{I} + \mathbf{J}$; good QNs $|F, m_F\rangle$. Strong field $g_J\mu_B B \gg A$: $\mathbf{I}, \mathbf{J}$ decouple; good QNs $|m_J, m_I\rangle$ (Paschen-Back of hyperfine).

(b) Cross-over field

Set $g_J\mu_B B_c \simeq A$:

$$A = h \cdot 1420 \text{ MHz} \simeq 9.41 \times 10^{-25} \text{ J}.$$

With $g_J \simeq 2$ for the $1^2S_{1/2}$ ground state:

$$B_c = \frac{A}{g_J\mu_B} = \frac{9.41 \times 10^{-25} \text{ J}}{2 \cdot 9.27 \times 10^{-24} \text{ J/T}}.$$

Evaluate:

$$\boxed{B_c \simeq 50 \text{ mT } (\sim 500 \text{ G}).} \quad (8)$$

(c) Weak-field eigenenergies

Total spin $\mathbf{F} = \mathbf{I} + \mathbf{J}$, $\mathbf{I}\cdot\mathbf{J} = \frac{\hbar^2}{2}[F(F+1) - I(I+1) - J(J+1)]$.

For $I = J = 1/2$, $F = 0, 1$:

$$E_{F=1}^{(0)} = +\frac{1}{4}A\hbar^2, \quad E_{F=0}^{(0)} = -\frac{3}{4}A\hbar^2.$$

Apply projection theorem within fixed F :

$$\langle F m_F | J_z | F m_F \rangle = \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)} m_F \hbar.$$

Read off the Landé g_F :

$$g_F = g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)} - \frac{m_e}{m_p} g_I \frac{F(F+1) + I(I+1) - J(J+1)}{2F(F+1)}.$$

Combine zero-field and Zeeman:

$$\boxed{E(F, m_F) = A\hbar^2 \frac{F(F+1) - I(I+1) - J(J+1)}{2} + g_F\mu_B B m_F \hbar.} \quad (9)$$

For ground-state H: $g_{F=1} \simeq +1$, $g_{F=0} = 0$ to leading order in m_e/m_p . Triplet splits linearly with B ; singlet unshifted at first order.

(d) Strong-field eigenenergies

In $|m_J, m_I\rangle$ Zeeman is diagonal. Decompose:

$$\mathbf{I}\cdot\mathbf{J} = I_z J_z + \frac{1}{2}(I_+ J_- + I_- J_+).$$

Drop ladder pieces (off-diagonal in the strong-field basis):

$$\langle m_J, m_I | \mathbf{I}\cdot\mathbf{J} | m_J, m_I \rangle = \hbar^2 m_J m_I.$$

Add the Zeeman terms:

$$E(m_J, m_I) = A\hbar^2 m_J m_I + g_J \mu_B B m_J \hbar - g_I \mu_N B m_I \hbar.$$

Insert the four sign combinations:

$$E_1 = E(+\frac{1}{2}, +\frac{1}{2}) = +\frac{1}{4}A\hbar^2 + \frac{1}{2}g_J \mu_B B \hbar - \frac{1}{2}g_I \mu_N B \hbar, \quad (10)$$

$$E_2 = E(+\frac{1}{2}, -\frac{1}{2}) = -\frac{1}{4}A\hbar^2 + \frac{1}{2}g_J \mu_B B \hbar + \frac{1}{2}g_I \mu_N B \hbar, \quad (11)$$

$$E_3 = E(-\frac{1}{2}, +\frac{1}{2}) = -\frac{1}{4}A\hbar^2 - \frac{1}{2}g_J \mu_B B \hbar - \frac{1}{2}g_I \mu_N B \hbar, \quad (12)$$

$$E_4 = E(-\frac{1}{2}, -\frac{1}{2}) = +\frac{1}{4}A\hbar^2 - \frac{1}{2}g_J \mu_B B \hbar + \frac{1}{2}g_I \mu_N B \hbar. \quad (13)$$

Stretched states $|\uparrow\uparrow\rangle, |\downarrow\downarrow\rangle$ ($|m_F| = 1$) are exact eigenstates at all B ; the $m_F = 0$ pair is given exactly by Breit–Rabi.

(e) Forces and the H maser

High-field upper states E_1, E_2 have $m_J = +1/2$; magnetic moment $\mu_z \simeq -\mu_B$ (electron dominates). Force:

$$F_z = -\frac{\partial E}{\partial z} = -(g_J \mu_B m_J \hbar) \frac{\partial B_z}{\partial z} \mp (\frac{1}{2}g_I \mu_N \hbar) \frac{\partial B_z}{\partial z}.$$

Take leading electronic piece, $g_J \mu_B/2 \simeq \mu_B$, $m_H \simeq m_p$:

$$a \simeq \frac{\mu_B}{m_p} \frac{\partial B_z}{\partial z} = \frac{9.27 \times 10^{-24} \text{ J/T}}{1.67 \times 10^{-27} \text{ kg}} \cdot 20 \text{ T/m}.$$

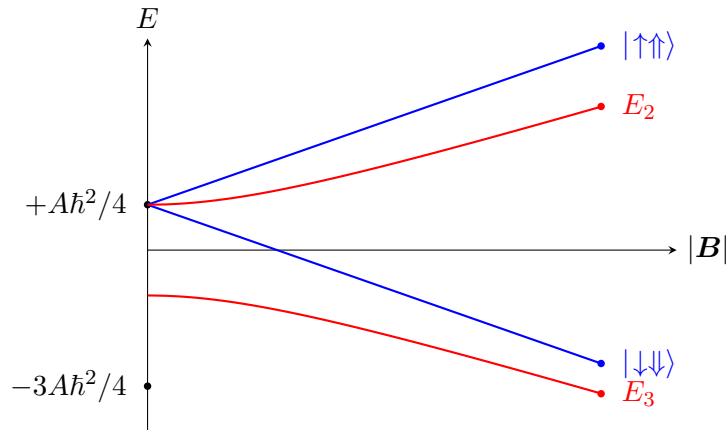
Evaluate:

$$|a| \simeq 1.1 \times 10^5 \text{ m s}^{-2} \text{ for both upper states.} \quad (14)$$

Fractional force difference set by nuclear Zeeman:

$$\frac{\Delta F}{\langle F \rangle} \simeq \frac{g_I \mu_N}{g_J \mu_B} \sim \frac{1}{1836} \sim 5 \times 10^{-4}.$$

Breit–Rabi connection. Stretched states $|\uparrow\uparrow\rangle, |\downarrow\downarrow\rangle$ correspond to $|F = 1, m_F = \pm 1\rangle$. The $m_F = 0$ pair: E_2 at high field connects to $|F = 1, m_F = 0\rangle$; E_3 connects to $|F = 0, m_F = 0\rangle$.



Hydrogen maser. The Stern–Gerlach gradient deflects low-field-seeking states E_1, E_2 ($m_J = +1/2$) into a PTFE-coated storage bulb. Inside a low-field microwave cavity tuned to $\nu_{\text{hf}} = 1420 \text{ MHz}$, only the upper $F = 1$ levels are populated, the $|F = 0, m_F = 0\rangle$ ground-hyperfine state is empty, and the $\Delta F = 1, \Delta m_F = 0$ M1 transition is driven by stimulated emission. Wall-collision lifetimes of $\sim \text{s}$ give an extremely narrow Lorentzian and a precision frequency standard.

Question 3 — Three- vs. four-level lasers and the Yb fibre

Problem. (a) Threshold contrast between three- and four-level systems. (b) Derive the gain coefficient from the Einstein relations. (c)–(g) Yb-doped fibre: $r = 2\text{ }\mu\text{m}$, $L = 0.5\text{ m}$, $\sigma_{\text{abs}} = 6 \times 10^{-25}\text{ m}^2$, $\sigma_{\text{em}} = 2.5 \times 10^{-25}\text{ m}^2$, 86% pump absorption at 910 nm, lower-band sublevels at $E_{1A}/hc = 60\,000\text{ m}^{-1}$, $E_{1B}/hc = 106\,000\text{ m}^{-1}$, upper level at $1\,026\,000\text{ m}^{-1}$.

(a) Threshold for three- vs. four-level lasers

Inversion $N^* = N_2 - (g_2/g_1)N_1$ must overcome cavity loss.

Four-level: $|1\rangle$ above the ground state, thermally empty ($N_1 \sim e^{-\Delta E/k_B T} \rightarrow 0$); inversion needs an infinitesimal fraction of ground population pumped.

Three-level: $|1\rangle$ is the ground state; need $N_2 > N_1$, i.e. $> \frac{1}{2}$ of all ions pumped. Threshold pump rate $\sim$ spontaneous-decay rate $\times N/2$, orders of magnitude larger.

(b) Gain coefficient

Rate equation for a beam of intensity I , lineshape $g(\nu - \nu_0)$, in the unsaturated regime:

$$\frac{dI}{dz} = \frac{\hbar\omega_0}{c} g(\nu - \nu_0) (B_{21}N_2 - B_{12}N_1) I.$$

Assumptions: (i) plane wave; (ii) homogeneous broadening, $\int g d\nu = 1$; (iii) populations not depleted; (iv) thermal photons negligible at optical frequency.

Define $\gamma(\nu)$ by $dI/dz = \gamma I$:

$$\gamma(\nu) = \frac{\hbar\omega_0}{c} g(\nu - \nu_0) (B_{21}N_2 - B_{12}N_1).$$

Apply $g_1 B_{12} = g_2 B_{21}$:

$$\gamma(\nu) = \frac{\hbar\omega_0}{c} g(\nu - \nu_0) B_{21} \left[N_2 - \frac{g_2}{g_1} N_1 \right].$$

Eliminate B_{21} via $A_{21} = \hbar\omega_0^3 B_{21}/(\pi^2 c^3)$:

$$\boxed{\gamma(\nu) = \frac{\pi^2 c^2 A_{21}}{\omega_0^2} g(\nu - \nu_0) \left[N_2 - \frac{g_2}{g_1} N_1 \right] = \sigma_{\text{em}}(\nu) N^*} \quad (15)$$

(c) Yb ion density

Beer absorption: $I(L)/I(0) = e^{-\sigma_{\text{abs}} N L} = 0.14$ (86% absorbed):

$$\sigma_{\text{abs}} N L = -\ln 0.14 = 1.97.$$

Solve for N :

$$N = \frac{1.97}{6 \times 10^{-25}\text{ m}^2 \cdot 0.5\text{ m}}.$$

Evaluate:

$$\boxed{N \simeq 6.6 \times 10^{24}\text{ m}^{-3}} \quad (16)$$

(d) Lower-band thermal populations at 300 K

Boltzmann factors with $g_0 = g_{1A} = g_{1B} = 1$. At $T = 300$ K, $k_B T / hc \simeq 2.084 \times 10^4 \text{ m}^{-1}$:

$$\frac{E_{1A}}{k_B T} = \frac{60\,000}{2.084 \times 10^4} \simeq 2.88, \quad \frac{E_{1B}}{k_B T} = \frac{106\,000}{2.084 \times 10^4} \simeq 5.09.$$

Partition sum:

$$Z_{\text{low}} = 1 + e^{-2.88} + e^{-5.09} = 1 + 0.0561 + 0.00614 \simeq 1.062.$$

Fractional populations $f_\alpha = e^{-E_\alpha/k_B T} / Z_{\text{low}}$:

$$\boxed{f_{1A} \simeq 5.3 \times 10^{-2}, \quad f_{1B} \simeq 5.8 \times 10^{-3}.} \quad (17)$$

(e) Threshold pump energies

Threshold: $N_{2,\text{th}} = N_{1\alpha} \simeq f_{1\alpha}(N - N_2) \simeq f_{1\alpha}N$ ($f_{1\alpha} \ll 1$).

Pump-photon energy:

$$E_{\text{pump}} = \frac{hc}{910 \times 10^{-9} \text{ m}} \simeq 2.18 \times 10^{-19} \text{ J} \simeq 1.36 \text{ eV}.$$

Core volume:

$$V = \pi r^2 L = \pi (2 \times 10^{-6} \text{ m})^2 \cdot 0.5 \text{ m} \simeq 6.28 \times 10^{-12} \text{ m}^3.$$

Total ions:

$$NV \simeq 6.6 \times 10^{24} \text{ m}^{-3} \cdot 6.28 \times 10^{-12} \text{ m}^3 \simeq 4.14 \times 10^{13}.$$

Threshold pump energy $E_{\alpha,\text{th}} = f_{1\alpha} N V E_{\text{pump}}$:

$$E_{A,\text{th}} = 5.3 \times 10^{-2} \cdot 4.14 \times 10^{13} \cdot 2.18 \times 10^{-19} \text{ J}.$$

$$E_{B,\text{th}} = 5.8 \times 10^{-3} \cdot 4.14 \times 10^{13} \cdot 2.18 \times 10^{-19} \text{ J}.$$

Evaluate:

$$\boxed{E_{A,\text{th}} \simeq 0.48 \text{ }\mu\text{J}, \quad E_{B,\text{th}} \simeq 52 \text{ nJ}.} \quad (18)$$

(f) Round-trip gain at $E = 5E_{B,\text{th}}$

Excited density:

$$N_2 = 5f_{1B}N = 5 \cdot 5.8 \times 10^{-3} \cdot 6.6 \times 10^{24} \text{ m}^{-3} \simeq 1.9 \times 10^{23} \text{ m}^{-3}.$$

Lower-band populations of remaining ions:

$$N_{1A} \simeq f_{1A}(N - N_2) \simeq 3.4 \times 10^{23} \text{ m}^{-3}, \quad N_{1B} \simeq f_{1B}(N - N_2) \simeq 3.7 \times 10^{22} \text{ m}^{-3}.$$

Inversions ($g_2/g_{1\alpha} = 1$):

$$N_A^* = 1.9 \times 10^{23} \text{ m}^{-3} - 3.4 \times 10^{23} \text{ m}^{-3} \simeq -1.5 \times 10^{23} \text{ m}^{-3} < 0.$$

$$N_B^* = 1.9 \times 10^{23} \text{ m}^{-3} - 3.7 \times 10^{22} \text{ m}^{-3} \simeq +1.5 \times 10^{23} \text{ m}^{-3}.$$

Round-trip gain on $2 \rightarrow 1B$:

$$G_{\text{rt}} = \exp(2\sigma_{\text{em}} N_B^* L) = \exp(2 \cdot 2.5 \times 10^{-25} \text{ m}^2 \cdot 1.5 \times 10^{23} \text{ m}^{-3} \cdot 0.5 \text{ m}).$$

Evaluate the exponent:

$$2\sigma_{\text{em}} N_B^* L = 3.75 \times 10^{-2}.$$

$$\boxed{G_{\text{rt}} \simeq e^{0.038} \simeq 1.04 (\sim 4\%); \text{ no gain on } 2 \rightarrow 1A.} \quad (19)$$

(g) Maximum efficiency and losses

Quantum-defect efficiency:

$$\eta_{\text{QD}} = \frac{\lambda_{\text{pump}}}{\lambda_{\text{laser}}} = \frac{E_{\text{laser}}}{E_{\text{pump}}}.$$

$$2 \rightarrow 1B: E_{\text{laser}}/hc = (1\,026\,000 - 106\,000) \text{ m}^{-1} = 920\,000 \text{ m}^{-1}, E_{\text{pump}}/hc = 1\,099\,000 \text{ m}^{-1}:$$

$$\eta_{\text{QD}}^B = \frac{920}{1099} \simeq 0.837.$$

$$2 \rightarrow 1A: E_{\text{laser}}/hc = 966\,000 \text{ m}^{-1}:$$

$$\eta_{\text{QD}}^A = \frac{966}{1099} \simeq 0.879.$$

Combine with $\eta_{\text{diode}} = 0.60$ and $\eta_{\text{abs}} = 0.86$:

$$\eta_{\text{max}}^B = 0.60 \cdot 0.86 \cdot 0.837 \simeq 0.43.$$

$$\eta_{\text{max}}^A \simeq 0.60 \cdot 0.86 \cdot 0.879 \simeq 0.45.$$

$$\boxed{\eta_{\text{max}} \simeq 43\text{--}45\%}. \quad (20)$$

Practical losses. Ground-state absorption on the laser line (especially $2 \rightarrow 1B$, quasi-three-level); excited-state absorption from level 2; non-radiative quenching by impurities or Yb clusters; cavity scattering; sub-optimal output coupling; pump-heating broadening of the emission line.

Question 4 — Radiative transfer and lithium absorption

Problem. Recast $dI_\nu/dz = -\alpha_\nu I_\nu + j_\nu$ in optical-depth form, solve for constant S_ν , and use measured ratios $I_A : I_B : I_C = 0.85 : 1 : 0.15$ to extract τ_ν and $S_\nu/I_\nu(0)$. Show $\sigma_{\text{abs}}^{\text{res}} = \lambda_0^2/(2\pi)$ for a lifetime-broadened line, evaluate for $\text{Li } 2s \rightarrow 2p$ at 671 nm, give the column density for $\tau \sim 1$ in a 1 cm cell, and discuss when natural-width lines are observed; list the source requirements for the $\text{Li } 2s \rightarrow np$ Rydberg series.

(a) Optical-depth form and formal solution

Define:

$$d\tau_\nu = \alpha_\nu dz, \quad S_\nu = \frac{j_\nu}{\alpha_\nu}.$$

Divide transfer equation by α_ν :

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu.$$

Multiply by integrating factor e^{τ_ν} :

$$\frac{d}{d\tau_\nu}(I_\nu e^{\tau_\nu}) = S_\nu e^{\tau_\nu}.$$

Integrate from 0 to τ_ν at constant S_ν :

$$I_\nu(\tau_\nu)e^{\tau_\nu} - I_\nu(0) = S_\nu(e^{\tau_\nu} - 1).$$

Solve for I_ν :

$$\boxed{I_\nu(\tau_\nu) = I_\nu(0)e^{-\tau_\nu} + S_\nu(1 - e^{-\tau_\nu})}. \quad (21)$$

Beer's law in the limit $\tau_\nu \rightarrow 0$; black-body S_ν in the limit $\tau_\nu \rightarrow \infty$.

(b) Inversion of the flame measurement

Identify the three measurements with (21):

$$I_A = I_\nu(0)e^{-\tau_\nu} + S_\nu(1 - e^{-\tau_\nu}), \quad I_B = I_\nu(0), \quad I_C = S_\nu(1 - e^{-\tau_\nu}).$$

Subtract I_C from I_A :

$$I_A - I_C = I_B e^{-\tau_\nu}.$$

Divide by I_B , set $X = I_A/I_B, Y = I_C/I_B$:

$$e^{-\tau_\nu} = X - Y = 0.85 - 0.15 = 0.70.$$

Take log:

$$\boxed{\tau_\nu = -\ln 0.70 \simeq 0.357.} \quad (22)$$

Read $S_\nu/I_\nu(0)$ from I_C :

$$\frac{S_\nu}{I_\nu(0)} = \frac{Y}{1 - e^{-\tau_\nu}} = \frac{0.15}{1 - 0.70}.$$

Evaluate:

$$\boxed{\frac{S_\nu}{I_\nu(0)} = 0.50.} \quad (23)$$

Optically thin flame; carbon emits with half the incoming black-body brightness at this frequency.

(c) Resonant cross section and the Li $2s \rightarrow 2p$ line

Lifetime-limited Lorentzian, $\Gamma = A_{21}$:

$$g_L(\omega - \omega_0) = \frac{1}{2\pi} \frac{\Gamma}{(\omega - \omega_0)^2 + (\Gamma/2)^2}.$$

Evaluate on resonance:

$$g_L(0) = \frac{2}{\pi A_{21}}.$$

Insert in $\sigma_{\text{abs}} = (g_2/g_1)(\pi^2 c^2 A_{21}/\omega_0^2) g_L$, take $g_2/g_1 = 1$:

$$\sigma_{\text{abs}}^{\text{res}} = \frac{\pi^2 c^2 A_{21}}{\omega_0^2} \cdot \frac{2}{\pi A_{21}} = \frac{2\pi c^2}{\omega_0^2}.$$

Use $\omega_0 = 2\pi c/\lambda_0$:

$$\boxed{\sigma_{\text{abs}}^{\text{res}} = \frac{\lambda_0^2}{2\pi} \text{ (independent of } A_{21}\text{).}} \quad (24)$$

For Li, $\lambda_0 = 671 \text{ nm}$:

$$\sigma_{\text{abs}}^{\text{res}} = \frac{(671 \times 10^{-9} \text{ m})^2}{2\pi} \simeq 7.2 \times 10^{-14} \text{ m}^2.$$

$$\boxed{\sigma_{\text{abs}}^{\text{res}} \simeq 7 \times 10^{-14} \text{ m}^2.} \quad (25)$$

Column density for $\tau \sim 1$ at $L = 1 \text{ cm}$:

$$N \sim \frac{1}{\sigma_{\text{abs}}^{\text{res}} L} = \frac{1}{7.2 \times 10^{-14} \text{ m}^2 \cdot 10^{-2} \text{ m}}.$$

Evaluate:

$$N_{\text{Li}} \sim 1.4 \times 10^{15} \text{ m}^{-3} (\sim 10^9 \text{ cm}^{-3}). \quad (26)$$

When are pure lifetime-broadened lines seen? Natural width must dominate Doppler ($\propto \sqrt{T/M}$) and pressure ($\propto N\sigma_{\text{coll}}\bar{v}$) widths: very low temperature (laser cooling, transverse atomic beams with $\mathbf{k} \cdot \mathbf{v} \rightarrow 0$), very low pressure ($\tau_{\text{coll}} \gg \tau_{\text{rad}}$), and absence of inhomogeneous mechanisms (Stark, transit-time). Examples: forbidden lines via saturated absorption, Ramsey spectroscopy, single-ion traps.

(d) Source for the $2s \rightarrow np$ Li series

Li ground state $1s^2 2s$, ionisation $5.39 \text{ eV} \equiv 43\,484 \text{ cm}^{-1}$, series limit $\lambda_{\text{lim}} \simeq 230 \text{ nm}$.

Required source properties:

(i) *Continuum coverage* from $\sim 671 \text{ nm}$ down to $\sim 230 \text{ nm}$: D_2 arc, Xe arc, tungsten + UV continuum, or synchrotron radiation, so absorption appears as dips on a smooth background.

(ii) *High UV brightness*: peak cross section $\propto 1/n^{*5}$ for high Rydberg states; need adequate UV signal-to-noise.

(iii) *Featureless/stable spectrum*: any source structure mimics absorption; alternative is a swept tunable laser of sufficient range.

(iv) *Resolving power* $\Delta\nu/\nu \lesssim 10^{-6}$ to resolve Doppler width at $T \sim 700 \text{ K}$: Echelle, Fourier-transform, or two-photon Doppler-free laser spectroscopy.

(v) *Long-term intensity stability*: faint high- n lines need steady I/I_0 ratios over long acquisition.

In short: continuum or broadly tunable laser covering 230–700 nm, high UV brightness, excellent stability, recorded at high resolving power against a wavelength reference.